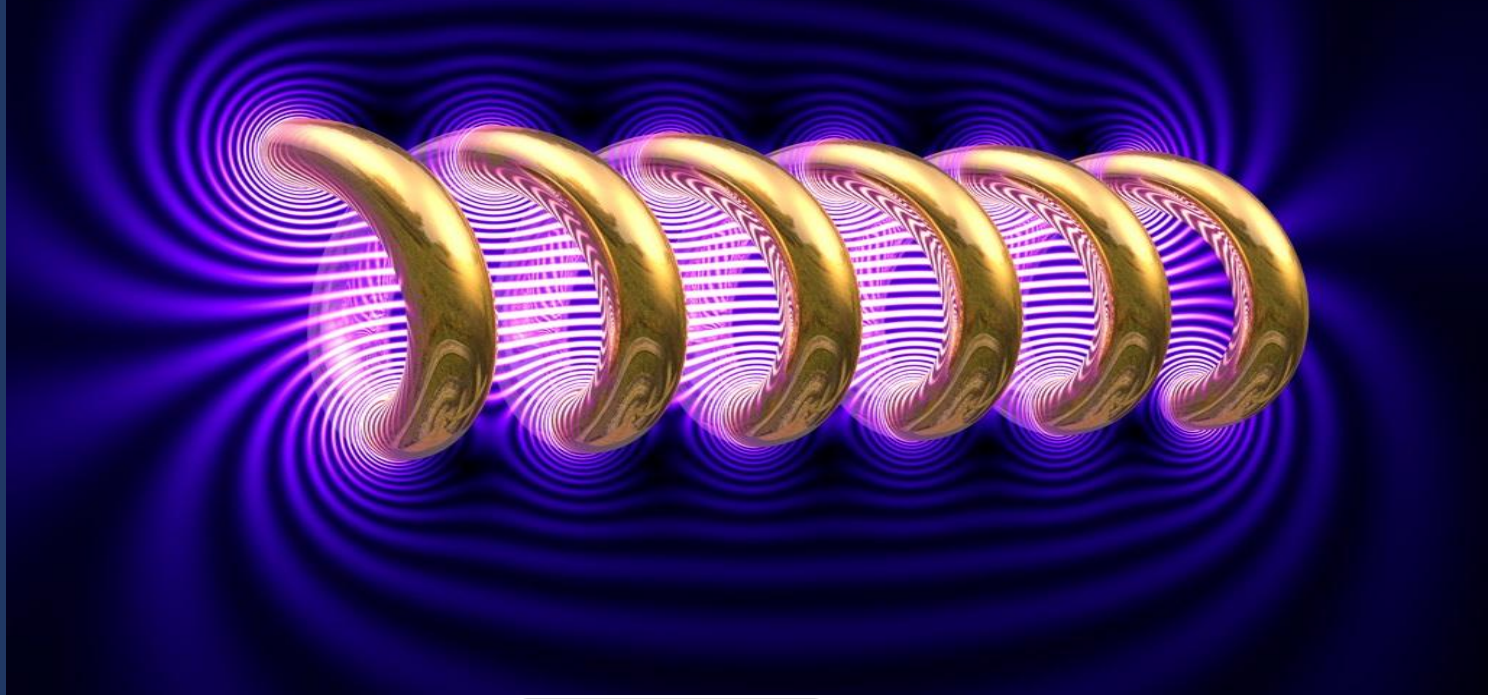


CHAPTER – 4

MOVING CHARGES AND MAGNETISM

Hans Christian Oersted observed that, when a compass is placed near a straight wire carrying a current, the compass needle aligns so that it is tangent to a circle drawn around the wire (neglecting the influence of the Earth's magnetic field on the compass). Oersted's discovery provided the first link between electricity and magnetism. The space or region around the current carrying conductor within which its influence can be felt by the magnetic needle is called the **magnetic field** of the current carrying conductor in general way. However, the definition of $\vec{B}$ can be given as the number of lines of magnetic field per unit area or the force acting on unit charge which is moving with unit velocity perpendicular to the magnetic field.

S.I. unit of magnetic field is $\mathrm{Wm^{-2}}$ or T (tesla).

The strength of magnetic field is called one tesla, if a charge of one coulomb, when moving with a velocity of $1\ \mathrm{ms^{-1}}$ along a direction perpendicular to the direction of the magnetic field experiences a force of one newton.

1 tesla (T) = 1 weber $\mathrm{metre^{-2}}$ ($\mathrm{Wb\ m^{-2}}$)

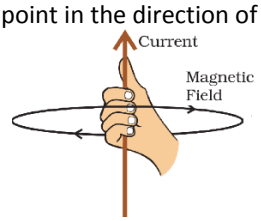
= 1 newton $\mathrm{ampere^{-1}\ metre^{-1}}$ ($\mathrm{N\ A^{-1}\ m^{-1}}$)

C.G.S. units of the magnetic field are called gauss or oersted.

1 gauss $= 10^{-4}$ tesla.

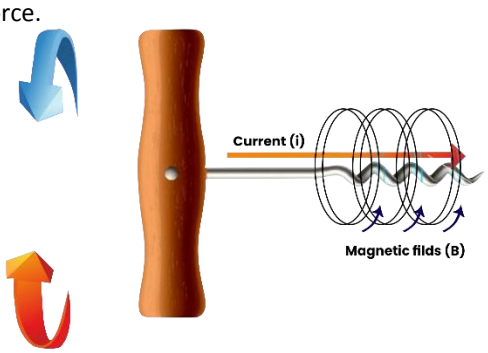
Right-hand thumb rule

If the linear conductor is grasped in the palm of the right hand with the thumb pointing along the direction of the current, then the curl fingers will point in the direction of lines of force.



Maxwell's Cork Screw Rule

If a right-handed cork screw is rotated so that its tip moves in the direction of flow of current through the conductor, then the rotation of the head of the screw gives the direction of magnetic lines of force.



The conventional sign for a magnetic field coming out of the plane and normal to it is a dot i.e. ⊙The magnetic field perpendicular to the plane in the down ward direction is denoted by ⓧ.

Magnetic force

A charge q inside an electric field of strength $\vec{E}$ experiences force $\vec{F}$, which is given by

$$|\vec{F}| = |q\vec{E}| = qE$$

Motion of a charge inside electric field

If a potential difference V is applied between two parallel plates, a uniform electric field is set up between the plates. Its strength is given by $E = V/d$

A charge q of mass m experiences force $F = q\,E$,

which produces an acceleration, $a = \frac{qE}{m}$

The charge entering an electric field into perpendicular direction or at an angle follows a parabolic path.

Lorentz force

The total force experienced by a charge moving inside the electric and magnetic fields is called Lorentz force. It is given by

$\vec{F} = q\,(\vec{E} + \vec{v} \times \cdot \vec{B})$

Magnetic force on a current-carrying conductor

Consider a rod of a uniform cross-sectional area A and length l. We shall assume one kind of mobile carriers as in a conductor (here electrons). Let the number density of these mobile charge carriers in it be n. Then the total number of mobile charge carriers in it is nlA. For a steady current I in this conducting rod, we may assume that each mobile carrier has an average drift velocity v_d. In the presence of an external magnetic field B, the force on these carriers is:

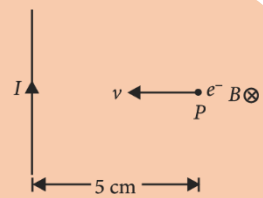
$F = (nlA)q\, v_d \times B$

where q is the value of the charge on a carrier. Now nqv_d is the current density j and $|(nq\, v_d)|A$ is the current I. Thus,

$F = [(nqv_d)l\, A] \times B = [\, jAl\,] \times B = Il \times B$

where l is a vector of magnitude l, the length of the rod, and with a direction identical to the current I.

Q. A long straight wire carries a certain current and produces a magnetic field 2×10^{-4} Wb m^{-2} at a perpendicular distance of 5 cm from the wire. An electron situated at 5 cm from the wire moves with a velocity 10^7 m/s towards the wire along perpendicular to it. Then find the force experienced by the electron. (Charge on electron 1.6×10^{-19} C)



Sol. The situation is as shown in the figure.

Here, $v = 10^7$ m/s,

$B = 2 \times 10^{-4}\ \mathrm{Wb/m^2}$

The magnitude of the force experienced by the electron is

$F = evB \sin\theta$

$\because \vec{v}$ and $\vec{B}$ are perpendicular to each other

$= evB \sin 90° = 1.6 \times 10^{-19} \times 10^7 \times 2 \times 10^{-4} \times 1 = 3.2 \times 10^{-16}$ N

Motion of charge inside magnetic field

A charge q of mass m moving with velocity $\vec{v}$ inside a magnetic field of strength $\vec{B}$ experiences force,

$|\vec{F}| = q\,|\vec{v} \times \vec{B}| = B\,q\,v \sin\theta$

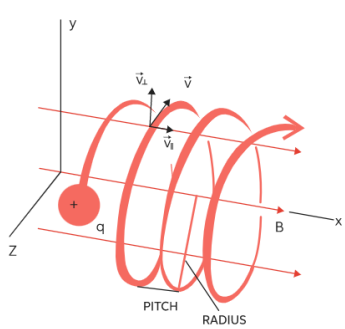
where θ is angle between the direction of motion of charge ($\vec{v}$) and the direction of magnetic field ($\vec{B}$). This force acts perpendicular to both $\vec{v}$ and $\vec{B}$ *i.e.,* the direction of motion of the charge and the direction of applied magnetic field.

The charge does not experience any force, if it is at rest or if it moves along the direction of magnetic field. The force is maximum, when charge moves perpendicular to the direction of magnetic field.

(i) If $\vec{v}$ and $\vec{B}$ are perpendicular ($\theta = 90°$), the force on the charged particle makes it to move along circular path, whose radius is given by

$r = \frac{mv}{Bq}$

(ii)



If $\vec{v}$ and $\vec{B}$ act at an angle θ, then due to the component of velocity $v \sin\theta$ (perpendicular to $\vec{B}$, the charge moves along circular path of radius r, which is given by)

$r = \frac{mv \sin\theta}{Bq}$

while due to the component of velocity $v\cos\theta$ (along $\vec{B}$), the charge at the same time moves along the direction of magnetic field. As a result, the charge moves along a helical path and pitch of the helical path is given by

pitch $\frac{2\pi\, mv\cos\theta}{Bq}$

Cyclotron

Cyclotron is a particle accelerator and is used to accelerate positive ions. Under the action of magnetic field, the positive ions move along spiral path and gain energy as they cross the alternating electric field again and again. Cyclotron is based on the principle that the positive ions can be accelerated to high energies with a comparatively smaller alternating potential difference by making them to cross the electric field again and again, by making use of a strong magnetic field. In a cyclotron, the frequency of applied alternating electric field is equal to frequency of oscillation of the positive ion and this frequency is called **cyclotron frequency.** It is given by

$\nu = \frac{Bq}{2\,\pi\, m}$

where m and q are mass and charge of the positive ion and B is strength of the magnetic field.

The positive ions of charge q and mass m in cyclotron attain maximum energy which is given by

(i) $E_{\max} = \frac{1}{2}.\frac{B^2 q^2 R^2}{m}$ where R is radius of the dees of the cyclotron.

(ii) $E_{\max} = 2n\,(V\,q)$.
where n is number of revolutions completed by the positive ions before leaving the dees.

Uses of Cyclotron

For several decades, these were the best sources of high-energy beams for nuclear physics experiments. However, these are still in use for this type of research.

Limitations of the cyclotron

(i) Cyclotron cannot accelerate uncharged particle like neutron.

(ii) The positively charged particles having large mass i.e. ions cannot move at limitless speed in a cyclotron.

Q. A 10 eV electron is circulating in a plane at right angles to a uniform field at magnetic induction 10^{-4} Wb/m^2 (= 1.0 gauss), then find the orbital radius of electron.

Sol. Kinetic energy of electron $\left(\frac{1}{2} \times mv^2\right) = 10$ eV and magnetic induction $(B) = 10^{-4}$ Wb/m^2

Therefore, $\frac{1}{2}(9.1 \times 10^{-31})v^2 = 10 \times (1.6 \times 10^{-19})$

or, $v^2 = \frac{2\times10\times(1.6\times10^{-19})}{9.1\times10^{-31}} = 3.52 \times 10^{12}$

or, $v = 1.876 \times 10^6$ m/s

Centripetal force, $\frac{mv^2}{r} = Bev$

Therefore, $r = \frac{mv}{Be} = \frac{(9.1\times10^{-31})\times(1.876\times10^6)}{10^{-4}\times(1.6\times10^{-19})}$

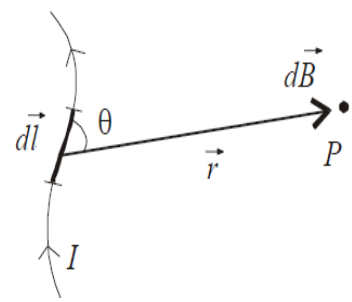
$= 11 \times 10^{-2}$ m $= 11$ cm

Biot savart's Law:

According to Biot savarts law, the magnetic field the is in magnitude due to a current element of length dl carrying a current I at a point at distance r from it, is given by

$$|\vec{dl}| = \frac{\mu_0 I}{4\pi}\cdot\frac{|\vec{dl} \times \hat{r}|}{r^2} = \frac{\mu_0}{4\pi}.\frac{I\,dl\sin\theta}{r^2}$$

where θ is the angle between the direction of the current and the line joining the current element to the point and μ_0 is absolute permeability of the free space ($\mu_0 = 4\pi \times 10^{-7}$ T A^{-1} m).



The direction of magnetic field $\vec{dB}$ is that of $I\,\vec{dl} \times \vec{r}$.

Q. Find the magnetic field $d\vec{B}$ due to a small current element $d\vec{l}$ at a distance $\vec{r}$ and element carrying current i.
According to Biot-Savart's law,

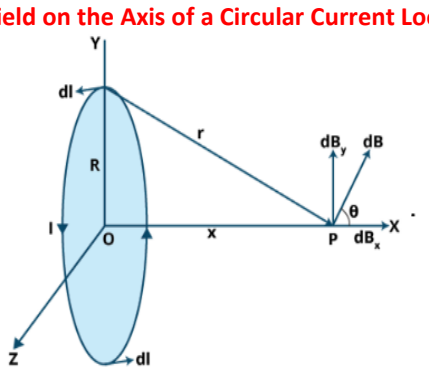
Sol. $d\vec{B} \propto i\left(\frac{d\vec{l}\times\hat{r}}{r^3}\right)$

We know that,

$\hat{r} = \frac{\vec{r}}{r}$

$d\vec{B} = \frac{\mu_0}{4\pi} i\left(\frac{d\vec{l}\times\vec{r}}{r^3}\right)$

Magnetic Field on the Axis of a Circular Current Loop



Let us consider a circular loop of radius R with I flowing through it. Let P be the point and its distance from center of the loop O be x. The magnetic field at point P is computed by taking two diametrically opposite line segments each of length $\vec{dl}$ at both ends of the wire, which is shown by the points from which the dl vector is taken in the diagram. The vector which is joining the current element at dl and P is $\hat{r}$.

According to the Biot-savart's law, the magnetic induction at point P because of the current element A is

$\vec{dB} = \frac{\mu_0}{4\pi}\frac{\vec{dl}\times\hat{r}}{r^2}$

Here, the direction of the $\vec{dB}$ is perpendicular to the current element I $\vec{dl}$ and AP. Consequently, it is along the direction of PQ. The magnitude of magnetic field induction $\vec{dB}$ can be written a

$dB = \frac{\mu_0}{4\pi} \frac{Idl\, sin\theta}{r^2}$

Here, θ is the angle between current element I $\overrightarrow{dl}$ and $\hat{r}$ Since its value is 90°, the magnitude of dB becomes:

$dB = \frac{\mu_0}{4\pi} \frac{Idl}{r^2}$

The magnetic field induction $\overrightarrow{dB}$ corresponding to each current element is resolved into two components dBcosθ and dBsinθ along the x-direction and y-direction respectively. The vertical components are equal and opposite in direction, hence they cancel each other whereas the horizontal components contribute to the total magnetic field. Hence, the net magnetic field B due to the entire circular loop at the point P is:

$\vec{B} = \int \overrightarrow{dB} = dBcos\theta\hat{\imath}$

$\vec{B} = \frac{\mu_0}{4\pi} \int \frac{Idl}{r^2}\, cos\theta\hat{\imath}$

By considering the triangle OAP as shown in the diagram,

$\cos\theta = \frac{R}{(x^2+R^2)^2}$

were, $r^2 = R^2 + x^2$

we get,

$\vec{B} = \frac{\mu_0}{4\pi} \frac{R}{(x^2+R^2)^{\frac{3}{2}}} \hat{\imath} \int dl$

Since it is a circular loop to calculate the net magnetic field at the point P, we integrate the line element from 0 to 2πR

$\vec{B} = \frac{\mu_0 I}{2} \frac{R^2}{(x^2+R^2)^{\frac{3}{2}}} \hat{\imath}$

If the circular loop contains N turns, then the magnetic field is given by

$\vec{B} = \frac{\mu_0 IN}{2} \frac{R^2}{(x^2+R^2)^{\frac{3}{2}}} \hat{\imath}$

Ampere's Circuital Law-

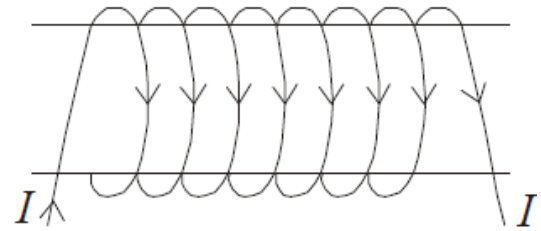
Statement- Ampere's circuital law states that the line integral of the magnetic field around any closed path in free space is equal to absolute permeability (μ_0) times the net current enclosed by the path.

$\oint \vec{B}.\overrightarrow{dl} = \mu_0 I$

Where, $\vec{B}$ is the magnetic field $\overrightarrow{dl}$ is small element, μ_0 is the absolute permeability of free space and *I* is the current.

The Solenoid

Ampere's circuital law holds good for a closed path of any size and shape around a current carrying conductor because the relation is independent of distance from conductor. A straight solenoid consists of hollow tube over which a large number of turns of insulated copper wire are uniformly wound.



If a solenoid of *n* turns per unit length carries a current *I*, then

(i) magnetic field at a point well inside the solenoid is given by

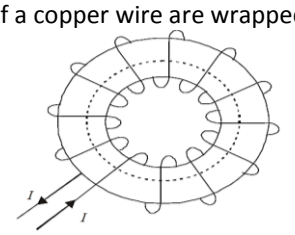
$$B = \mu_0\, n\, I$$

(ii) magnetic field at a point on one end of the solenoid is given by

$B = \frac{1}{2}\mu_0\, nI$

The Toroid

A toroidal solenoid is an anchor ring around which is large number of turns of a copper wire are wrapped.



The magnetic field produced in toroid will be same at all points on the circumference of the circle and at any point it will act along the tangent to the ring

$B = \frac{\mu_0 NI}{2\pi r}$

For any point inside the empty space surrounded by the toroid and outside the toroid, magnetic field is zero because the net current enclosed in this space is zero.

Magnetic field intensity at a point outside the cylinder (*r* > *a*)

$B = \frac{\mu_0 I}{2\pi r}$

The magnetic field intensity at a point outside the cylinder varies inversely as the distance of the point from the axis of the cylinder.

Q. A long solenoid of 50 cm length having 100 turns carries a current of 2.5 A. Then find the magnetic field at the centre of the solenoid. ($\mu_0 = 4\pi \times 10^{-7}$ T m A^{-1})

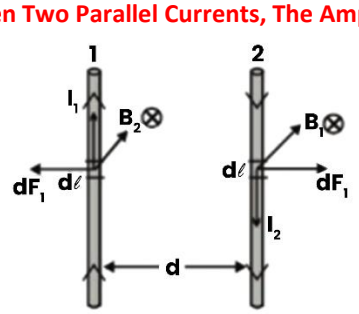
Sol. Here, l = 50 cm, N = 100, i = 2.5 A

$B = \mu_0 ni = \frac{\mu_0 NI}{l}$

Magnetic field inside the solenoid,

$B = \frac{4\pi\times10^{-7}\times100\times2.5}{0.5} = 6.28 \times 10^{-4}$ T

Force Between Two Parallel Currents, The Ampere



When two infinitely long parallel conductors carrying currents I_1 and I_2 are placed a distance *r* apart, then force on the unit length of a conductor due to the other conductor is given by

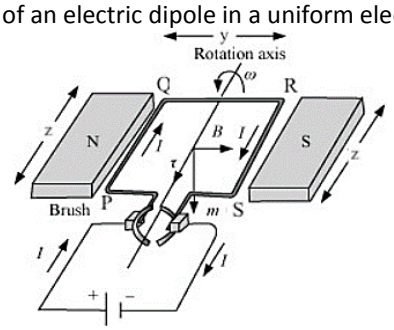
$F = \frac{\mu_0}{4\pi} . \frac{2\, I_1\, I_2}{r}$

The force is attractive, if currents in two conductors is in same direction; and repulsive, if currents are in opposite directions.

If the currents in both parallel wires are equal and in same direction then magnetic field at a point exactly half way between the wire is zero.

Torque on a current carrying coil placed inside a magnetic field

A magnetic dipole is the limit of either a closed loop of electric current or a pair of poles as the dimensions of the source are reduced to zero while keeping the magnetic moment constant. Now we shall show that a steady current I passing through a rectangular loop placed in a uniform magnetic field experiences a torque. It does not experience a net force. This behavior is similar to the of an electric dipole in a uniform electric field.



Let's consider a case when the rectangular loop is placed such that the uniform magnetic field B is the plane of the loop. The field exerts no force on both arms PS and QR of the loop. It is perpendicular to the arm PQ of the loop and exerts a force F_1 on it which is directed into the plane of the loop. Its magnitude is,

$F_1 = IzB$

Similarly, it exerts a force F_2 on the arm RS and F_2 is directed out of the plane of the paper.

$F_2 = IzB = F_1$

Therefore, the net force on the loop is zero. As both the forces F_1 and F_2 nullify each other, there is a torque on the loop. Here, we can see that the torque on the loop tends to rotate it in an anti-clockwise direction.

$\tau = F_1 (y/2) + F_2 (y/2)$

$= IzB (y/2) + IzB (y/2)$

$= I (y \times z) B$

$= IAB$ .... eq(1)

where $A = y \times z$ is the area of the rectangle.

Now consider a case when the plane of the loop is not along the magnetic field but makes an angle with it. And also consider the angle between the field and the normal to the coil is angle Θ. The forces on both the arms QR and SP are equal, opposite and act along the axis of the coil, which connects the centers of mass of QR and SP. Being collinear along the axis they cancel out each other, resulting in no net force or torque. The forces on arms PQ and RS are F_1 and F_2. Furthermore, they too are equal and opposite, with magnitude,

$F_1 = F_2 = IzB$

As they are not collinear it results in a couple. The effect of torque is, however, less than the earlier case when the plane of the loop was along the magnetic field. The magnitude of the torque on the loop is,

$\tau = F_1 (y/2) \sin\theta + F_2 (y/2) \sin\theta$

$= I (y \times z) B \sin\theta$

$= IAB\sin\theta$ ....eq(2)

So, the torques in equations (1) and (2) can be expressed as the vector product of the magnetic moment of the coil and the magnetic field. Therefore, we can define the magnetic moment of the current loop as,

$m = IA$

where A is the direction of the area vector. The angle between m and B is θ, the equations (1) and (2) can be expressed by one expression

$\tau = m \times B$

where m is the magnetic moment and B is the uniform magnetic field.

Circular current loop as a magnetic dipole

An arrangement of two equal and opposite magnetic poles separated by a small distance is called a magnetic dipole.

The magnetic dipole moment of a magnetic dipole is defined as the product of its pole strength and magnetic length. It is a vector quantity, directed from S-pole to N-pole.

Ampere found that the distribution of magnetic lines of force around a finite current carrying solenoid is similar to that produced by a bar magnet.

The magnetic induction at a point along the axis of a circular coil carrying current is

$$\boldsymbol{B} = \frac{\mu_0 n I a^2}{2(a^2+x^2)^{\frac{3}{2}}}$$

The direction of this magnetic field is along the axis and is given by right hand rule. For points which are far away from the centre of the coil, *x>>a*, a^2 is small and it is neglected. Hence for such points,

$$B = \frac{\mu_0 n I a^2}{2x^3}$$

If we consider a circular loop, n = 1, its area $A = \pi a^2$

$$B = \frac{\mu_0 I A}{2\pi x^3}$$

The magnetic induction at a point along the axial line of a short bar magnet is

$$B = \frac{\mu_0 2M}{4\pi x^3}$$

$$B = \frac{\mu_0 M}{2\pi x^3} \quad \text{.....eq(2)}$$

By comparing equations (1) and equation (2), we find that,

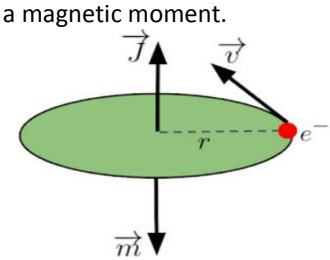
M=IA

Hence a current loop is equivalent to a **magnetic dipole of moment M = IA**

The magnetic moment of a current loop is defined as the product of the current and the loop area. Its direction is perpendicular to the plane of the loop.

The magnetic dipole moment of a revolving electron

An electron revolving in an orbit about the nucleus of an atom behaves like a current carrying loop. It is called a minute current-loop and produces a magnetic field. Every current loop is associated with a magnetic moment.



Let us consider that the magnetic moment associated with a loop carrying current I having area A.

μ_L=i.A

The current due to a revolving electron is,

$$i = \frac{e}{T}$$

Where, T= the period of revolution of electron motion around the nucleus that is

$T = \frac{2\pi r}{v}$

e= Charge on an electron So from the above equation

$i = \frac{\frac{e}{2\pi r}}{v}$

$i = \frac{ev}{2\pi r}$

The area of the current loop is:

$A = \pi r^2$

Now put the value of I and A in equation, μ_L= i . A

$\mu_L = \left(\frac{ev}{2\pi r}\right)\pi r^2$

$\mu_L = \frac{evr}{2}$

$\mu_L = \frac{evr}{2}\left(\frac{m}{m}\right)$

$\mu_L = \frac{e}{2m}(\text{mvr})$

$$\mu_L = \frac{e}{2m}L$$

Where L= mvr=The orbital angular momentum of the electron

Another value of L is,

$L = \frac{nh}{2\pi}$

$\mu_L = \frac{e}{2m}\frac{nh}{2\pi}$

$\mu_L = n\frac{eh}{4m\pi}$

Where n=1,2, 3…… is the principal quantum number.

This equation gives the magnetic moment associated with the orbital motion of the electron.

Q. A rectangular coil of length 0.12 m and width 0.1 m having 50 turns of wire is suspended vertically in a uniform magnetic field of strength 0.2 Weber/m^2. The given coil carries a current of 2 A. If the plane of the coil is inclined at an angle of 30° with the direction of the field, then find torque required to keep the coil in stable equilibrium.

Sol. The required torque is $\tau = NIA\,B\sin\theta$

where N is the number of turns in the coil, I is the current through the coil, B is the uniform magnetic field, A is the area of the coil and q is the angle between the direction of the magnetic field and normal to the plane of the coil.

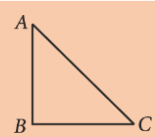
Here, $N = 50, I = 2\text{ A}, A = 0.12\text{ m} \times 0.1\text{ m} = 0.012\text{ m}^2$

$B = 0.2\text{ Wb/m}^2$ and $\theta = 90° - 30° = 60°$

$\therefore \quad \tau = (50)(2\text{ A})(0.012\text{ m}^2)(0.2\text{ Wb/m}^2)\sin 60°$

= 0.20 N m

Q. A current carrying closed loop in the form of a right-angle isosceles triangle ABC is placed in a uniform magnetic field acting along AB. If the magnetic force on the arm BC is $\vec{F}$ then find the force on the arm AC.



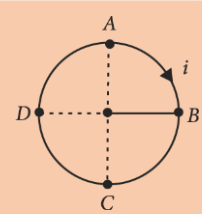
Sol. Here, $\vec{F}_{BC} = \vec{F}$ and $\vec{F}_{AB} = I(\vec{l} \times \vec{B}) = 0$

The net magnetic force on a current carrying closed loop in a uniform magnetic field is zero.

$\therefore \quad \vec{F}_{AB} + \vec{F}_{BC} + \vec{F}_{AC} = 0$

$\Rightarrow \quad \vec{F}_{AC} = -\vec{F}_{BC} \quad (\because \vec{F}_{AB} = 0)$

Q. A circular coil $ABCD$ carrying a current 'i' is placed in a uniform magnetic field. If the magnetic force on the segment AB is $\vec{F}$, then what will be the force on the remaining segment $BCDA$



Sol. The net magnetic force on a current loop in a uniform magnetic field is always zero.

$\therefore \quad \vec{F}_{AB} + \vec{F}_{BCDA} = 0$

$\vec{F}_{BCDA} = -\vec{F}_{AB} = -\vec{F}$

Q. A circular loop of area 0.01 m^2 carrying a current of 10 A, is held perpendicular to a magnetic field of intensity 0.1 T. Then find the torque acting on the loop.

Sol. Area (A) = 0.01 m^2; Current (I) = 10 A;

Angle (ϕ) = 90° and magnetic field (B) = 0.1 T

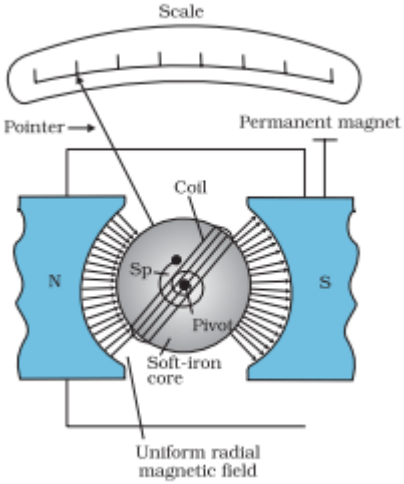
Therefore, actual angle $\theta = (90° - \phi)$

$= (90° - 90°) = 0°$

Torque acting on the loop (τ) = $IAB\sin\theta$

$= 10 \times 0.01 \times 0.1 \times \sin 0° = 0$

The Moving Coil Galvanometer-



A moving coil galvanometer is a device used to detect flow of current in a circuit. A moving coil galvanometer consists of a rectangular coil placed in a uniform radial magnetic field produced by cylindrical pole pieces. Torque on coil τ = NIAB where N is number of turns, A is area of coil. If C is torsional rigidity of material of suspension wire, then for deflection θ, torque τ = Cθ

∴ For equilibrium NIAB = Cθ

$\theta = \frac{NAB}{C} I \Rightarrow \theta \alpha I$

Clearly, deflection in galvanometer is directly proportional to current, so the scale of galvanometer is linear.

Figure of Merit of a galvanometer

The current which produces a deflection of one scale division in the galvanometer is called its figure of Merit. It is equal to $\frac{I}{\theta} + \frac{C}{NAB}$

Sensitivity of a galvanometer

Current sensitivity: It is defined as the deflection of coil per unit current flowing in it

Sensitivity $S_I = \frac{\theta}{I} = \frac{NAB}{C}$

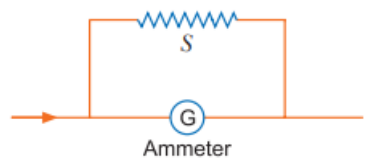
Voltage sensitivity

It is defined as the deflection of coil per unit potential difference across its ends.

$S_V = \frac{\theta}{V} = \frac{NAB}{R_g.C'}$

Where, R_g is resistance of galvanometer. Clearly for greater sensitivity, number of turns N, area A and magnetic field strength B should be large and torsional rigidity C of suspension should be small.

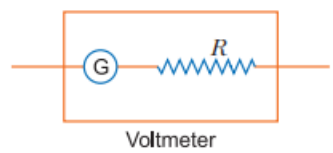
Conversion of Galvanometer into Ammeter



A galvanometer can be converted into ammeter by using very small resistance in parallel with the galvanometer coil. The small resistance connected in parallel is called a shunt. If G is resistance of galvanometer, I_g is current in galvanometer for full scale deflection, then for conversion of galvanometer into ammeter of range I ampere, the shunt is given by

$S = \frac{I_g}{I - I_g} G$

Conversion of Galvanometer into Voltmeter



A galvanometer may be converted into voltmeter by connecting high resistance (R) in series with the coil of galvanometer. If V volt is the range of voltmeter formed, then series resistance is given by

$R = \frac{V}{I_g} - G$

Q. Current sensitivity of a moving coil galvanometer is 5 div/mA and its voltage sensitivity (angular deflection per unit voltage applied) is 20 div/V. The resistance of the galvanometer is

Sol. Let N = number of turns in galvanometer,
A = Area, B = magnetic field
k = the restoring torque per unit twist.

Current sensitivity, $I_S = \frac{NBA}{k}$

Voltage sensitivity, $V_S = \frac{NBA}{kR_G}$

So, resistance of galvanometer

$R_G = \frac{I_S}{V_S} = \frac{5 \times 1}{20 \times 10^{-3}} = \frac{5000}{20} = 250\ \Omega$

Q. A milli voltmeter of 25 milli volt range is to be converted into an ammeter of 25 ampere range. Then find the value (in ohm) of necessary shunt.

Sol. $S = \frac{V_g}{(I - I_g)}$ By Neglecting I_g we get,

$\therefore \quad S = \frac{V_g}{I} = \frac{25 \times 10^{-3}\ \text{V}}{25\ \text{A}} = 0.001\ \Omega$

SUMMARY

- **Force on a Straight Conductor:**
  Force F on a straight conductor of length *l* and carrying a steady current I placed in a uniform external magnetic field B,
  $\vec{F} = I\vec{l}x\vec{B}$
- **Lorentz Force:**
  Force on a charge q moving with velocity v in the presence of magnetic and electric fields B and E.
  $\vec{F}=q(\vec{v}x\vec{B}+\vec{E})$
- **Magnetic Force:**
  The magnetic force $\vec{F}_B = q(\vec{v}x\vec{B})$ is normal to $\vec{V}$ and work done by it is zero.
- **Cyclotron:**
  A charge q executes a circular orbit in a plane normal with frequency called the cyclotron frequency given by,
  $vc = \frac{qB}{2\pi m}$
  This cyclotron frequency is independent of the particle's speed and radius.
- **Biot – Savart Law:**
  It asserts that the magnetic field $d\vec{B}$ due to an element $d\vec{l}$ carrying a steady current I at a point P at a distance r from the current element is,
  $d\vec{B} = \frac{\mu_0}{4\Pi} I \frac{d\vec{l}x\vec{r}}{r^3}$
- **Magnetic Field due to Circular Coil:**
  Magnetic field due to circular coil of radius R carrying a current I at an axial distance X from the centre is
  $B = \frac{\mu_0 I R^2}{2(X^2+R^2)^{3/2}}$
  At the centre of the coil,
  $B = \frac{\mu_0 I}{2R}$
- **Ampere's Circuital Law:**
  For an open surface S bounded by a loop C, then the Ampere's law states that
  $\int_c^o \vec{B}.d\vec{l} = \mu_0 I$
  where I refer to the current passing through S.
- If B is directed along the tangent to every point on the perimeter them
  $BL = \mu_0 I_e$
  Where I_e is the net current enclosed by the closed circuit.
- **Magnetic Field:**
  Magnetic field at a distance R from a long, straight wire carrying a current I is given by,
  $B = \frac{\mu_0 I}{2R}$
  The field lines are circles concentric with the wire.
- **Magnetic field B inside a long Solenoid carrying a current I:**
  $B = \mu_0 nI$
  Where n is the number of turns per unit length.
- For a toroid,
  $B = \frac{\mu_0 NI}{2\prod r}$
  Where N is the total numbers of turns and r is the average radius.
- **Magnetic Moment of a Planar Loop:**
  Magnetic moment m of a planar loop carrying a current I, having N closely wound turns, and an area A, is
  $\vec{m} = NI\vec{A}$
- **Direction of $\vec{m}$ is given by the Right – Hand Thumb Rule:**
  Curl and palm of your right hand along the loop with the fingers pointing in the direction of the current, the thumb sticking out gives the direction of
  $\vec{m}\ (and\ \vec{A})$
- **Loop placed in a Uniform Magnetic Field:**
  (a) When this loop is placed in a uniform magnetic field B, Then, the force F on it is, F = 0
  And the torque on it is, $\vec{\tau} = \vec{m}x\vec{B}$
  In a moving coil galvanometer, this torque is balanced by (a) counter torque due to a spring, yielding.
  kΦ = NI AB
  where Φ is the equilibrium deflection and k the torsion constant of the spring.
- **Magnetic Moment in an Electron:**
  An electron moving around the central nucleus has a magnetic moment μ_l, given by
  $\mu_l = \frac{e}{2m} l$
  where *l* is the magnitude of the angular momentum of the circulating electron about the central nucleus.
- **Bohr Magneton:**
  The smallest value of μ_l is called the Bohr magneton μ_B
  $\mu_B = 9.27 \times 10^{-24}$ J/T

MIND MAP

Moving Charges & Magnetism

Magnetic Field $(\vec{B})$

- It is a region around a magnet or current carrying conductor or a moving charge in which its magnetic effect can be felt
- SI unit is Tesla(T) = weber/m^2
- 1 Gauss = 10^{-4} Tesla

Magnetic Force on moving charge

$\vec{F} = q\vec{v}\times\vec{B}$

$= qvB\sin\theta$

- For $\theta = 0$, $\vec{F} = 0$ along the magnetic field
- For $\theta = 90°$, i.e. if charge's velocity is perpendicular to field direction, Force is perpendicular to both field & velocity

$F = qvB = \dfrac{mv^2}{r}$

$r = \dfrac{mv}{qB}$ = radius of the circle in which charge rotates

Time period (T) = $\dfrac{2\pi m}{qB}$

ν(frequency) = $\dfrac{1}{T} = \dfrac{qB}{2\pi m}$

If $\theta \neq 0, 180°, 90°$

Then, $F = qvB\sin\theta$

And the charge particle will follow helix path whose

$r = \dfrac{m\,v_\perp}{q\,B}$ and pitch $= V_\parallel \times T = V_\parallel \times \dfrac{2\pi m}{qB}$

Biot-Savart Law

$\odot \quad \otimes$

$d\vec{B} = \dfrac{1}{4\pi\varepsilon_0 c^2}\dfrac{i d\vec{l}\times\vec{r}}{r^3}$

$d\vec{B} = \dfrac{\mu_0}{4\pi}\dfrac{i d\vec{l}\times\vec{r}}{r^3}$ $\quad\theta$ = angle between $d\vec{l}$ and $\vec{r}$

$dB = \dfrac{\mu_0}{4\pi}\dfrac{i dl\sin\theta}{r^2}$ $\quad$ Direction of field will be perpendicular to plane containing current element and the point of observation.

where $\mu_0 = \dfrac{1}{\varepsilon_0 c^2}$

$= 4\pi \times 10^{-7}\ \text{TmA}^{-1}$

Oersted Law

In April, 1820, Hans Christian Oersted discovered that flow of current in a wire could deflect a nearby magnetic compass needle

Ampere's Law

$\oint \vec{B}.d\vec{l} = \mu_0 i$

where i = total current crossing the area bounded by closed curve.

Solenoid

- Magnetic filed at a point inside due to a long solenoid $B = \mu_0 ni$
- And at point on one end $B = \dfrac{\mu_0 ni}{2}$

where n = no. of turns per unit length along the length of solenoid.

Magnetic field due to Toroid

$B = \dfrac{\mu_0 Ni}{2\pi r}$, N = total no. of turns, i = current in toroid

Force on a current carrying conductor

$d\vec{F} = i\,d\vec{l}\times\vec{B}$, $F = ilB$

Force between parallel currents

$F = \dfrac{\mu_0 i_1 i_2}{2\pi d}$

Torque experienced by a loop in uniform magnetic field

$\vec{\tau} = MB\sin\theta\,\hat{n} = \vec{M}\times\vec{B}$

Definition of Ampere

If two parallel wire kept 1 m apart, if $F = 2\times10^{-7}$, then current = 1A in each.

Sensitivity of moving coil galvanometer is nBA/k

Gal.to ammeter

$S = \dfrac{I_g}{I_T - I_g}G$

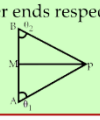
Gal.to voltameter

$R = \dfrac{V}{I_g} - G$

Field due to a straight wire current

$B = \dfrac{\mu_0 i}{4\pi d}(\cos\theta_1 - \cos\theta_2)$

where θ_1 and θ_2 are the angle corresponding to the lower and upper ends respectively



i.e $\theta_1 = 0$

$\theta_2 = \pi$

$B = \dfrac{\mu_0 i}{2\pi d}$

Field due to a circular current

Field at the centre

$B = \dfrac{\mu_0 i}{2a}$ $\quad \otimes \quad \odot$

Field at an axial point

$B = \dfrac{\mu_0 i\,a^2}{2(a^2+d^2)^{3/2}}$

Field at a point far away from the centre

i.e. for d>>>a

$B = \dfrac{\mu_0 i\,a^2}{2\,d^3}$

PRACTICE EXERCISE

MCQ

Q1. A portion of a conductive wire is bent in the form of a semicircle of radius r as shown below in figure. At the centre of this semicircle, the magnetic induction will be-

(a) zero (b) infinite
(c) $\frac{\mu_0}{4\mu}.\frac{5\pi i}{r}$ (d) $\frac{\mu_0}{4\mu}.\frac{\pi i}{r}$

Q2. A closely wound solenoid of 2000 turns and area of cross-section 1.5×10^{-4} m^2 carries a current of 2.0 *A*. It suspended through its center and perpendicular to its length, allowing it to turn in a horizontal plane in a uniform magnetic field 5×10^{-2} tesla making an angle of 30° with the axis of the solenoid. The torque on the solenoid will be-
(a) 3×10^{-2} N-m (b) 4.5×10^{-3} N-m
(c) 5.5×10^{-3} N-m (d) 1.5×10^{-2} N-m

Q3. An alternating electric field, of frequency *v*, is applied across the dees (radius = *R*) of a cyclotron that is being used to accelerate protons (mass = *m*). The operating magnetic field (B) used in the cyclotron and the kinetic energy (*K*) of the proton beam, produced by it
(a) $B = \frac{mv}{e}$ and $K = 2m\pi^2v^2R^2$
(b) $B = \frac{2\pi mv}{e}$ and $K = m^2\pi vR^2$
(c) $B = \frac{2\pi mv}{e}$ and $K = 2m\pi^2v^2R^2$
(d) $B = \frac{mv}{e}$ and $K = m^2\pi vR^2$

Q4. A galvanometer of 50-ohm resistance that has 25 divisions. A current of 4×10^{-4} ampere gives a deflection of one per division. To convert this galvanometer into a voltmeter having a range of 25 volts, it should be connected with a resistance of
(a) 2450 Ω in series (b) 2650 Ω in series
(c) 2500 Ω in series (d) 2750 Ω in series

Q5. If we double the radius of a coil keeping the current through it unchanged, then the magnetic field at any point at a large distance from the center becomes approximately
(a) double (b) five times
(c) four times (d) one-fourth

Q6. A particle of mass m, charge Q and kinetic energy T enters a transverse uniform magnetic field of induction $\vec{B}$. After 3 seconds, the kinetic energy of the particle will be:
(a) 3T (b) 2T
(c) T (d) 4T

Q7. A 10-eV electron is circulating in a plane at right angles to a uniform field at magnetic induction 10^{-4} Wb/m^2 (=1.0 gauss). The orbital radius of the electron is
(a) 12 cm (b) 16 cm
(c) 11 cm (d) 18 cm

Q8. A uniform electric field and a uniform magnetic field exist in a region in the same direction. An electron is projected with velocity pointed in the same direction. The electron will
(a) turn to its right
(b) turn to its left
(c) keep moving in the same direction but it Its speed will increase
(d) keep moving in the same direction but its speed will decrease

Q9. A moving coil galvanometer has 150 equal divisions. Its current sensitivity is 10-divisions per milliampere and voltage sensitivity is 2 divisions per millivolt. In order that each division reads 1 volt, the resistance in ohms needed to be connected in series with the coil will be
(a) 10^5 (b) 10^3
(c) 9995 (d) 99995

Q10. A 2 μC charge moving around a circle with a frequency of 6.25×10^{12} Hz produces a magnetic field 6.28 tesla at the center of the circle. The radius of the circle is
(a) 2.25m (b) 0.25 m
(c) 13.0 m (d) 1.25 m

Q11. A charged particle with charge *q* enters a region of constant, uniform and mutually orthogonal fields $\vec{E}$ and $\vec{B}$ with a velocity $\vec{v}$ perpendicular to both $\vec{E}$ and $\vec{B}$, and comes out without any change in magnitude or direction of $\vec{v}$. Then
(a) $\vec{v} = \vec{B} \times \vec{E}/E^2$ (b) $\vec{v} = \vec{E} \times \vec{B}/B^2$
(c) $\vec{v} = \vec{B} \times \vec{E}/B^2$ (d) $\vec{v} = \vec{E} \times \vec{B}/E^2$

Q12. A straight section *PQ* of a circuit lies along the *X*-axis from $x = -\frac{a}{2}$ to $x = \frac{a}{2}$ and carries a steady current *i*. The magnetic field due to the section *PQ* at a point *X* = + *a* will be
(a) proportional to *a* (b) proportional to a^2
(c) proportional to 1 /*a* (d) zero

Q13. A current *I* flow in an infinitely long wire with cross section in the form of a semi-circular ring of radius *R*. The magnitude of the magnetic induction along its axis is:
(a) $\frac{\mu_0 I}{2\pi^2 R}$ (b) $\frac{\mu_0 I}{2\pi R}$
(c) $\frac{\mu_0 I}{4\pi R}$ (d) $\frac{\mu_0 I}{\pi^2 R}$

Q14. A long solenoid carrying a current produces a magnetic field *B* along its axis. If the current is double and the number of turns per cm is halved, the new value of the magnetic field is
(a) 4*B* (b) *B*/2
(c) *B* (d) 2*B*

Q15. A particle of charge q and mass m moves in a circular orbit of radius r with angular speed ω. The radio of the magnitude of its magnetic moment to that of its angular momentum depends on

(a) ω and q (b) ω, q and m
(c) q and m (d) ω and m

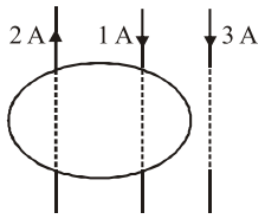
Q16. A current loop in a magnetic field

(a) can be in equilibrium in one orientation
(b) can be in equilibrium in two orientations, both the equilibrium states are unstable
(c) can be in equilibrium in two orientations, one stable while the other is unstable
(d) experiences a torque whether the field is uniform or non-uniform in all orientations

Q17. Two long parallel wires P and Q are held perpendicular to the plane of paper with distance of 5 m between them. If P and Q carry current of 2.5 amp. And 5 amp. Respectively in the same direction, then the magnetic field at a point half-way between the wires is

(a) $\mu_0/17$ (b) $\sqrt{3}\mu_0/2\pi$
(c) $\mu_0/2\pi$ (d) $3\mu_0/2\pi$

Q18. Two wires with currents 2 A and 1 A are enclosed in a circular loop. Another wire with current 3 A is situated outside the loop as shown. The $\oint \vec{B}.d\vec{l}$ around the loop is



(a) μ_0 (b) $3\mu_0$
(c) $6\mu_0$ (d) $2\mu_0$

Q19. If in a circular coil *A* of radius *R*, current *I* is flowing and in another coil *B* of radius 2*R* a current 2*I* is flowing, then the ratio of the magnetic fields B_A and B_B, produced by them will be

(a) 1 (b) 2
(c) 1/2 (d) 4

Q20. A charged particle moves through a magnetic field perpendicular to its direction. Then

(a) kinetic energy changes but the momentum is constant
(b) the momentum changes but the kinetic energy is constant
(c) both momentum and kinetic energy of the particle are not constant
(d) both momentum and kinetic energy of the particle are constant

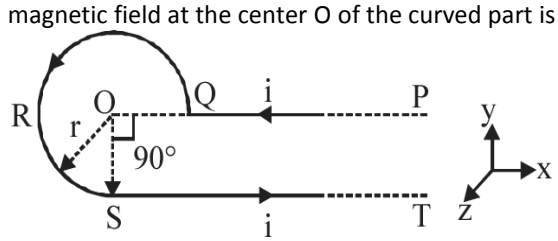
Q21. The magnitude of magnetic field at a point having perpendicular distance 50 mm from a long straight conducting wire carrying a current of 3 A is

(a) 0.12 G (b) 1.2 G
(c) 12 G (d) 0.012 G

Q22. A circular arc of wire of radius of curvature r subtends an angle of $\pi/4$ radian at its centre. If i current is flowing in it then the magnetic induction at its centre is –

(a) $\frac{\mu_0 i}{8r}$ (b) $\frac{\mu_0 i}{4r}$
(c) $\frac{\mu_0 i}{16r}$ (d) 0

Q23. A current i is flowing in a conductor PQRST shaped as shown in the figure. The radius of curved part QRS is r and length of straight portions PQ and ST is very large. The magnetic field at the center O of the curved part is –



(a) $\frac{\mu_0 i}{4\pi r}\left[\frac{3\pi}{2}+1\right]\hat{k}$ (b) $\frac{\mu_0 i}{4\pi r}\left[\frac{3\pi}{2}-1\right]\hat{k}$
(c) $\frac{\mu_0 i}{4\pi r}\left[\frac{3\pi}{2}+1\right](-\hat{k})$ (d) $\frac{\mu_0 i}{4\pi r}\left[\frac{3\pi}{2}-1\right](-\hat{k})$

Q24. The concept of displacement current was introduced by _____.

(a) Newton (b) Ampere
(c) Maxwell (d) Fleming

Q25. Lorentz force is given by the formula

(a) F= q(v + B + E) (b) F= q(v – B – E)
(c) F= q(v * B * E) (d) F= q(v * B + E)

ASSERTION AND REASONING

Directions: These questions consist of two statements, each printed as Assertion and Reason. While answering these questions, you are required to choose any one of the following four responses.

(a) If both Assertion and Reason are correct and the Reason is a correct explanation of the Assertion.
(b) If both Assertion and Reason are correct but Reason is not a correct explanation of the Assertion.
(c) If the Assertion is correct but Reason is incorrect.
(d) If both the Assertion and Reason are incorrect.

Q1. Assertion: To acerated positive ions, Cyclotron can be used.
Reason: Cyclotron frequency depends upon the velocity.

Q2. Assertion: Cyclotron does not accelerate electron.
Reason: Mass of the electrons is very small.

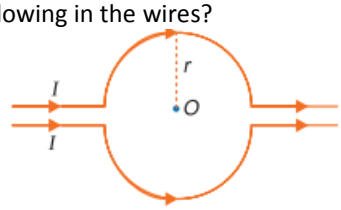
Q3. Assertion: In electric circuits, wires carrying currents in opposite directions are often twisted together
Reason: If the wires are not twisted together, the combination of the wires forms a current loop, the magnetic field generated by the loop might affect adjacent circuits or components.

Q4. Assertion: A charge, whether stationary or in motion produces a magnetic field around it.
Reason: Moving charges produce only electric field in the surrounding space.

S5. Assertion: If the current in a solenoid is reversed in direction while keeping the same magnitude, the magnetic field energy stored in the solenoid decreases.
Reason: Magnetic field energy density is proportional to square of current. If the current in a solenoid is reversed in direction while keeping the same magnitude, the magnetic field energy stored in the solenoid decreases.

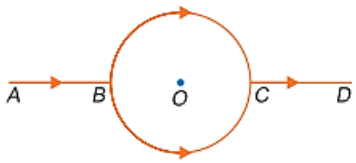
VERY SHORT ANSWER QUESTION

Q1. What is the value of magnetic field at point O due to current flowing in the wires?



Q2. An electron, passing through a region is not deflected. Are you sure that there is no magnetic field in that region?

Q3. What is the magnetic field at point O due to current carrying wires shown in figure?

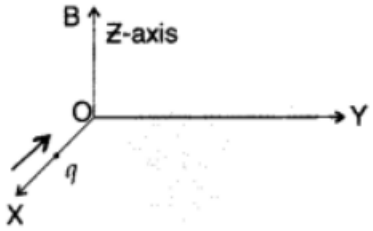


Q4. Magnetic field lines can be entirely confined within the core of a toroid, but not within a straight solenoid. Why?

Q5. Why do the electrostatic field lines not form closed loops?

SHORT ANSWER QUESTION

Q1. charge 'q' moving B along the X-axis with a velocity v is subjected to a uniform magnetic field B acting along the Z-axis as it crosses the origin O.
(i) Trace its trajectory.
(ii) Does the charge gain kinetic energy as it enters the magnetic field? Justify your answer.



Q2. State the underlying principle of a cyclotron. Write briefly how this machine is used to accelerate charged particles to high energies.

Q3. State two reasons why a galvanometer cannot be used as such to measure current in a given circuit.

Q4. Write any two important points of similarities and differences each between Coulomb's law for the electrostatic field and Biot-Savart's law for the magnetic field.

Q5. (a) In what respect is a toroid different from a solenoid? Draw and compare the pattern of the magnetic field lines in the two cases. (b) How is the magnetic field inside a given solenoid made strong?

NUMERICAL TYPE QUESTIONS

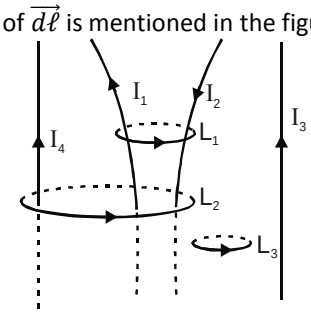
Q1. A metallic rod of mass per unit length 0.5 kg m^{-1} is lying horizontally on a smooth inclined plane which makes an angle of 30° with the horizontal. The rod is not allowed to slide down by flowing a current through it when a magnetic field of induction 0.25 T is acting on it in the vertical direction. Then determine the current flowing in the rod to keep it stationary.

Q2. A straight wire of length 0.5 metre and carrying a current of 1.2 ampere is placed in uniform magnetic field of induction 2 tesla. The magnetic field is perpendicular to the length of the wire. Then determine the force on the wire.

Q3. A charge having e/m equal to 10^8 C/kg and with velocity 3×10^5 m/s enters into a uniform magnetic field B = 0.3 tesla at an angle 30° with direction of field. Then determine the radius of curvature.

Q4. The resistance of an ammeter is 13 Ω and its scale is graduated for a current upto 100 amps. After an additional shunt has been connected to this ammeter it becomes possible to measure currents upto 750 amperes by this meter. Find the value of shunt resistance.

Q5. Find the values of $\oint \vec{B} \cdot \overrightarrow{d\ell}$ for the loops L_1, L_2, L_3 in the figure shown.
The sense of $\overrightarrow{d\ell}$ is mentioned in the figure.



HOMEWORK EXERCISE

MCQ

Q1. 1 Gauss =
(a) 10^4 Tesla (b) 10^{-4} Tesla
(c) 10^2 Tesla (d) 10^{-2} Tesla

Q2. Cyclotron is a device used to _____.
(a) Slow down charged particles (b) Accelerate the positively charged particles
(c) Stop the charged particles (d) None of the options

Q3. Magnetic field at any point inside the straight solenoid is given as———-
(a) $B = \mu_0 + nI$ (b) $B = \mu_0 + n + I$
(c) $B = \mu_0/nI$ (d) $B = \mu_0 nI$

Q4. When the charged particles move in a combined magnetic and electric field, then the force acting is known as _____.
(a) Centripetal force (b) Centrifugal force
(c) Lorentz force (d) Orbital force

Q6. SI unit of the magnetic field is _____.
(a) Dyne (b) Ohm
(c) Tesla (d) Volt

Q7. The magnetic moment of a current I carrying a circular coil of radius r and number of turns N varies as
(a) r^4 (b) r
(c) $1/r^4$ (d) r

Q8. An electron is projected with uniform velocity along the axis of a current carrying long solenoid. Which of the following is true?
(a) The electron will be accelerated along the axis
(b) The electron path will be circular about the axis
(c) The electron will experience a force at 45 degree to the axis and hence execute a helical path
(d) The electron will continue to move with uniform velocity along the axis of the solenoid

Q9. A current carrying closed loop of an irregular shape lying in more than one plane when placed in uniform magnetic field, the force acting on it
(a) will be more in the plane where its larger position is covered
(b) is zero
(c) is infinite
(d) may or may not be zero

Q10. A current loop placed in a non- uniform magnetic field experiences
(a) A force of repulsion
(b) A force of attraction
(c) A torque but not force
(d) A force and a torque

ASSERTION AND REASONING

Directions: These questions consist of two statements, each printed as Assertion and Reason. While answering these questions, you are required to choose any one of the following four responses.
(a) If both Assertion and Reason are correct and the Reason is a correct explanation of the Assertion.
(b) If both Assertion and Reason are correct but Reason is not a correct explanation of the Assertion.
(c) If the Assertion is correct but Reason is incorrect.
(d) If both the Assertion and Reason are incorrect.

Q1. Assertion: To convert a galvanometer into an ammeter a small resistance is connected in parallel with it.
Reason: The small resistance increases the combined resistance of the combination.

Q2. Assertion: We cannot think of magnetic field configuration with three poles.
Reason: A bar magnet does exert a torque on itself due to its own field.

Q3. Assertion: Magnetic moment of an atom is due to both, the orbital motion and spin motion of every electron.
Reason: A charged particle produces a magnetic field.

Q4. Assertion: A compass needle when placed on the magnetic north pole of the earth rotates in vertical direction.
Reason: The earth has only horizontal component of its magnetic field at the north pole

Q5. Assertion: The ferromagnetic substance do not obey Curies law.
Reason: At Curie point a ferromagnetic substance start behaving as a paramagnetic substance.

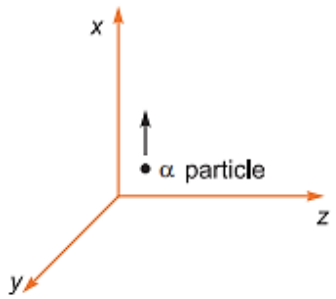
VERY SHORT ANSWER QUESTIONS

Q1. What can be the cause of helical motion of a charged particle?

Q2. Magnetic field lines can be entirely confined within the core of a toroid, but not within a straight solenoid. Why?

Q3. An electron does not suffer any deflection while passing through a region of uniform magnetic field. What is the direction of the magnetic field?

Q4. Why should an ammeter have a low resistance? A beam of a particles projected along + x-axis, experiences a force due to a magnetic field along the + y-axis. What is the direction of the magnetic field?



Q5. Why should an ammeter have a low resistance?

NUMERICAL TYPE QUESTIONS

Q1. A charged particle of mass 5 mg and charge q = +2mC has velocity $\vec{v} = 2\hat{\imath} - 3\hat{\jmath} + 4\hat{k}$. Find out the magnetic force on the charged particle and its acceleration at this instant due to magnetic field $\vec{B} = 3\hat{\jmath} - 2\hat{k}$. $\vec{v}$ and $\vec{B}$ are in m/s and Wb/m^2 respectively.

Q2. A charged particle has acceleration $\vec{a} = 2\hat{\imath} + x\hat{\jmath}$ in a magnetic field $\vec{B} = -3\hat{\imath} + 2\hat{\jmath} - 4\hat{k}$. Find the value of x.

Q3. A solenoid has a core of a material with relative permeability 400. The windings of the solenoid are insulated from the core and carry a current of 2A. If the number of turns is 1000 per meter, calculate
(a) H
(b) M
(c) B
(d) the magnetizing current I_m.

Q4. The core of toroid of 3000 turns has inner and outer radii of 11 cm and 12 cm respectively. A current of 0.6 A produces a magnetic field of 2.5 T in the core. Compute relative permeability of the core. ($\mu_0 = 4p \times 10^{-7}$ T m A^{-1}).

Q5. A magnetizing field of 1600 Am^{-1} produces a magnetic flux of 2.4×10^{-5} wb in an iron bar of cross-sectional area 0.2cm^2. Calculate permeability and susceptibility of the bar.

PRACTICE EXERCISE SOLUTIONS

MCQ

S1. (d) Here the straight part will does not contribute in magnetic field at the centre of the semicircle an it is because every element of the straight apart will be 0° or 180° with the line joining the centre and the element.

Due to circular portion, the field can be written as

$\frac{1}{2}\frac{\mu_0 i}{2r} = \frac{\mu_0 i}{4r}$

Hence total field at O $= \frac{\mu_0 i}{4r}$ tesla

S2. (d) Torque on the solenoid is given by the equation

$\tau = MB \sin\theta$

where θ is the angle between the magnetic field and the axis of solenoid.

M = niA

☐ τ = M B sin 30°

$= 2000 \times 2 \times 1.5 \times 10^{-4} \times 5 \times 10^{-2} \times \frac{1}{2}$

$= 1.5 \times 10^{-2}\ N\ m$

S3. (c) Time period of cyclotron can be given as,

$T = \frac{1}{v} = \frac{2\pi m}{eB}; B = \frac{2\pi m}{e}v; R = \frac{mv}{eB} = \frac{p}{eB}$

$\Rightarrow p = eBR = e \times \frac{2\pi m v}{e} R = 2\pi m v R$

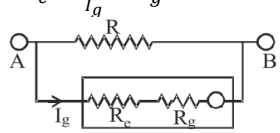
Kinetic Energy $= \frac{p^2}{2m} = \frac{(2\pi m v R)^2}{2m} = 2\pi^2 m v^2 R^2$

S4. (a) $R_g = 50\Omega$, $I_g = 25 \times 4 \times 10^{-A}\Omega = 10^{-2}$ A

Range of V = 25 volts

$V = Ig(R_e + R_g)$

$\therefore R_e = \frac{V}{I_g} - R_g = 2450\Omega$



S5. (c) $B_{\text{axis}} = \left(\frac{\mu_0 NI}{2x^3}\right) R^2$

$B \propto R^2$

So, when radius is doubled, magnetic field becomes four times.

S6. (c) When a charged particle enters transverse magnetic field it traverse a circular path. Its kinetic energy remains constant.

S7. (c) K.E. of electron = 10 eV

$\Rightarrow \frac{1}{2}\text{mv}^2 = 10\text{eV}$

$\Rightarrow \frac{1}{2}(9.1 \times 10^{-31})v^2 = 10 \times 1.6 \times 10^{19}$

$\Rightarrow v^2 = \frac{2\times10\times1.6\times10^{-19}}{9.1\times10^{-31}}$

$\Rightarrow v$

$\Rightarrow v^2 = 3.52 \times 10^{12} \Rightarrow v = 1.88 \times 10^6 m$

Also, we know that for circular motion

$\frac{mv^2}{r} = Bev \Rightarrow r = \frac{mv}{Be} = 11cm$

S8. (d) No magnetic force acts on the electron and force due to electric field will act opposite to its initial direction of motion. Hence its velocity decreases in magnitude.

S9. (c) Resistance of Galvanometer,

$G = \frac{\text{Current sensitivity}}{\text{Voltage sensitivity}} \Rightarrow G = \frac{10}{2} = 5\Omega$

Here i_g = Full scale deflection current

$= \frac{150}{10} = 15\text{mA}$

V = voltage to be measured = 150 volts (such that each division reads 1 volt)

$\Rightarrow R = \frac{150}{15\times10^{-3}} - 5 = 9995\Omega$

S10. (d) Magnetic field at the centre of the current loop is

$B = \frac{\mu_0 2\pi I}{4\pi R}$

or, $B = \frac{\mu_0 2\pi q v}{4\pi R}, R = \frac{\mu_0 2\pi q v}{4\pi B}$

Substituting the given values, we get

$R = \frac{4\pi\times10^{-7}\times2\pi\times10^{-6}\times6.25\times10^{12}}{4\pi\times6.28}$

=1.25m

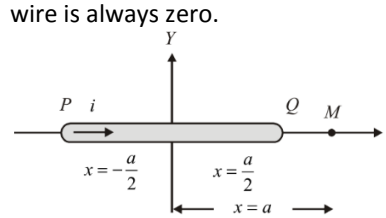
S11. (b) Here, $\vec{E}$ and $\vec{B}$are perpendicular to each other and the velocity $\vec{v}$does not change; therefore

$qE = qvB \Rightarrow v = \frac{E}{B}$

Also,

$$\left|\frac{\vec{E}\times\vec{B}}{B^2}\right| = \frac{EB\sin\theta}{B^2} = \frac{EB\sin 90°}{B^2} = \frac{E}{B} = |\vec{v}| = v$$

S12. (d) Magnetic field at a point on the axis of current carrying wire is always zero.

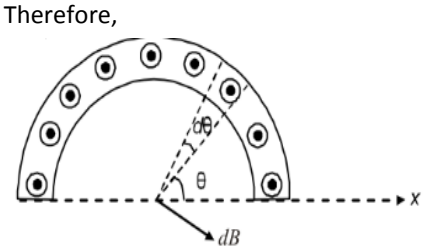


S13. (d) Current in a small element, $dI = \frac{d\theta}{\pi} I$Magnetic field due to the element

$dB = \frac{\mu_0}{4\pi}\frac{2dI}{R}$

The component dB cos θ, of the field is cancelled by another opposite component.

Therefore,



$$B_{net} = \int dB\sin\theta = \frac{\mu_0 I}{2\pi^2 R_0}\int_0^{\mu}\sin\theta\, d\theta = \frac{\mu_0 I}{\pi^2 R}$$

S14. (c) $B = \mu_0 ni$

$B_1 = (\mu_0)\left(\frac{n}{2}\right)(2i) = \mu_0 ni = B$

$\Rightarrow B_1 = B$

S15. (c) The angular momentum L of the particle is given by $L = mr^2\omega$ where ω = 2πn.

$\therefore$ Frequency$n = \frac{\omega}{2\pi}$; Further $i = q \times n = \frac{\omega q}{2\pi}$

Magnetic moment, $M = iA = \frac{\omega q}{2\pi} \times \pi r^2$;

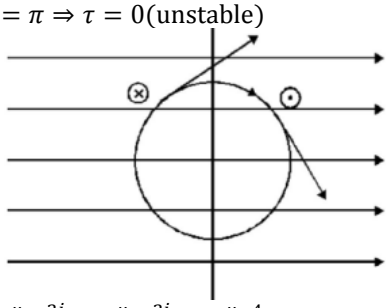
$\therefore M = \frac{\omega q r^2}{2}$ So, $\frac{M}{L} = \frac{\omega q r^2}{2 m r^2 \omega} = \frac{q}{2m}$

S16. (c) A current loop in a magnetic field is in equilibrium in two orientations one is stable and another unstable.

$\because \vec{\tau} = \vec{M} \times \vec{B} = MB\sin\theta$

If $\theta = 0° \Rightarrow \tau = 0$(stable)

If $\theta = \pi \Rightarrow \tau = 0$(unstable)



S17. (c) $B = \frac{\mu_0}{4\pi}\frac{2i_2}{(r/2)} - \frac{\mu_0}{4\pi}\frac{2i_1}{(r/2)} = \frac{\mu_0}{4\pi}\frac{4}{r}(i_2 - i_1)$

$= \frac{\mu_0}{4\pi}\frac{4}{5}(5 - 2.5) = \frac{\mu_0}{2\pi}$.

S18. (a) According to Ampere's circuit law

$\oint \vec{B}.\mathrm{d}\vec{l} = \mu_0 I_{\text{enclosed}} = \mu_0(2A - 1A) = \mu_0$

S19. (a) We know that the magnetic field produced by a current carrying circular coil of radius *r* at its centre is

$B = \frac{\mu_0}{4\pi}\frac{I}{r} \times 2\pi$

Here $B_A = \frac{\mu_0}{4\pi}\frac{I}{R} \times 2\pi$

and $B_B = \frac{\mu_0}{4\pi}\frac{2I}{2R} \times 2\pi$

$\Rightarrow \frac{B_A}{B_B} = 1$

S20. (b) When a charged particle enters a magnetic field at a direction perpendicular to the direction of motion, the path of the motion is circular. In circular motion the direction of velocity changes at every point (the magnitude remains constant).

Therefore, the tangential momentum will change at every point. But kinetic energy will remain constant as it is given by $\frac{1}{2}mv^2$ and v^2 is the square of the magnitude of velocity which does not change.

S21. (a) We know magnetic field due to a long straight current carrying wire

$B = \frac{\mu_0 i}{2\pi r} = \frac{4\pi \times 10^{-7} \times 3}{2\pi \times 50 \times 10^{-3}}$

(Note that $\mu_0 = 4\pi \times 10^{-7}$ in SI system)

= 1.20×10^{-5} Tesla = 0.12 G.

[As 1 Gauss = 10^{-4} Tesla]

S22. (c) The magnetic induction produced due to a current carrying arc at its centre of curvature is

$B = \frac{\mu_0 i \alpha}{4\pi r}$ ...(a)

(Subtending angle α at the centre of curvature)

$\Rightarrow \quad B = \frac{\mu_0 i \pi}{4\pi r} \times \frac{\pi}{4} = \frac{\mu_0 i}{16 r}$

S23 (a) $\vec{B}_O = \vec{B}_{QRS} + \vec{B}_{ST}$.

$\vec{B}_{PQ} = \text{zero}, \vec{B}_{QRS} = \frac{3}{4} \times \frac{\mu_0 i}{2r}\hat{k}, \vec{B}_{ST} = \frac{\mu_0 i}{4\pi r}\hat{k}$

$\Rightarrow \vec{B}_O = \frac{\mu_0 i}{4\pi r}\hat{k} + \frac{3}{4}\frac{3\mu_0 i}{2r}\hat{k} = \frac{\mu_0 i}{4\pi r}\left[\frac{3\pi}{2} + 1\right]\hat{k}$

S24. (c) Maxwell

The concept of displacement current was introduced by Maxwell.

S25. (d) Lorentz force is given by the formula F= q(v * B + E)

ASSERTION AND REASONING

S1. (c) Cyclotron is utilized to accelerate the positive ion. And cyclotron frequency is given by v=Be/2πm.

It means cyclotron frequency doesn't depend upon velocity. Therefore, assertion is true and reason false.

S2. (a) Cyclotron does not accelerate electron because the mass of electron is very small. It gets accelerated very appreciably as a result of which its mass increases. It results is mismatch between frequency of alternating current used and frequency of rotation of electron in the Dee's. So, cyclotron stops accelerating electrons after some time.

S3. (a) If the wires are twisted together, they can be formed as a single wire carrying currents in opposite directions. In this pattern, in wires no magnetic field is induced which does not affect adjacent circuits.

S4. (d) Assertion is incorrect, because only moving charges produce magnetic field around them, not stationary ones. Reason is incorrect, because moving charges produce both electric and magnetic field around them.

S5. (d) Reversing the direction of the current reverses the direction of the magnetic field. However, it has no effect on the magnetic-field energy density, which is proportional to the square of the magnitude of the magnetic field.

VERY SHORT ANSWER QUESTION

S1. Zero, because the upper and lower current carrying conductors are identical and so the magnetic fields caused by them at the center O will be equal and opposite.

S2. No, if an electron enters parallel to a magnetic field, no force acts and the electron remains undeflected.

S3. The magnetic field due to straight wires AB and CD is zero since either θ = 0° or 180° and that due to a semi-circular arc are equal and opposite; hence net field at O is zero.

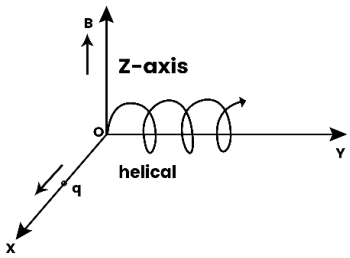
S4. At the edges of the solenoid, the field lines get diverged due to other fields and/or non-availability of dipole loops, while in toroid's the dipoles (in loops) orient continuously.

S5. Electric field lines do not form closed loops because the direction of an electric field is from positive to negative charge. So, one can regard a line of force

starting from a positive charge and ending on a negative charge. This indicates that electric field lines do not form closed loops.

SHORT ANSWER QUESTION

S1. (i)



(ii) Kinetic energy does not irrespective of the direction of the charge as,

Power delivered, $\vec{F}.\vec{v} = q(\vec{v} \times \vec{B}).\vec{v} = 0$

Scalar triple product $(\vec{v} \times \vec{B}).\vec{v} = 0$

S2. A cyclotron makes use of the principle that the energy of the charged particles can be increased to a high value by making it pass through an electric field repeatedly. The magnetic field acts on the charged particle and makes them move in a circular path inside the dee. Every time the particle moves from one dee to another it is acted upon by the alternating electric field, and is accelerated by this field, which increases the energy of the particle.

Uses: (i) It is used to bombard nuclei with high energetic particles accelerated by cyclotron and study the resulting nuclear reaction. (ii) It is used to implant ions into solids and modify their properties or even synthesize new materials.

S3. A galvanometer cannot be used as such to measure current due to following two reasons.

(i) A galvanometer has a finite large resistance and is connected in series in the circuit, so it will increase the resistance of circuit and hence change the value of current in the circuit.

(ii) A galvanometer is a very sensitive device, it gives a full-scale deflection for the current of the order of microampere, hence if connected as such it will not measure current of the order of ampere.

S4. Similarities: Both electrostatic field and magnetic field:

(i) follows the principle of superposition.

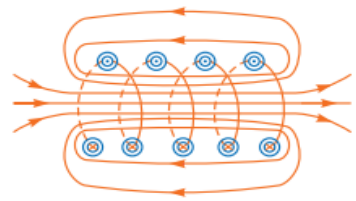
(ii) (ii) depends inversely on the square of distance from source to the point of interest.

Differences:

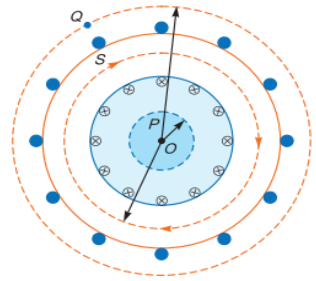
(i) Electrostatic field is produced by a scalar source (q) and the magnetic field is produced by a vector source ($I\vec{dl}$).

(ii) Electrostatic field is along the displacement vector between source and point of interest; while magnetic field is perpendicular to the plane, containing the displacement vector and vector source. (iii) Electrostatic field is angle independent, while magnetic field is angle dependent between source vector and displacement vector.

S5. (a) A toroid is a solenoid bent into the form of a closed ring. The magnetic field lines of solenoid are straight lines parallel to the axis inside the solenoid.



Solenoid



Toroid

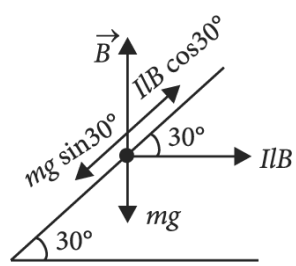
(b) The magnetic field lines of toroid are circular having common center. Inside a given solenoid, the magnetic field may be made strong by

(i) passing large current and

(ii) using laminated coil of soft iron.

NUMERICAL

S1.



Mass per unit length of a metallic rod is $\frac{m}{l} = 0.5\ \text{kg m}^{-1}$. Let I be the current flowing.

For equilibrium,

$mg \sin 30° = IlB \cos 30°$

$I = \frac{mg}{lB} \tan 30° = \frac{0.5 \times 9.8}{0.25 \times \sqrt{3}} = 11.32\ \text{A}$

S2. From, $F = il \times B = 1.2 \times 0.5 \times 2 = 1.2$ N

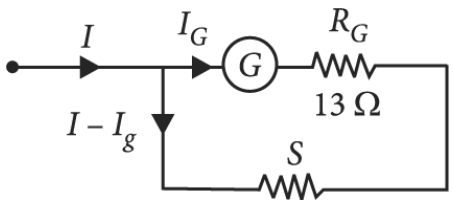
S3. $qvB \sin\theta = \frac{mv^2}{R}$

$R = \frac{mv}{qB \sin\theta} = \frac{3 \times 10^5}{10^8 \times 0.3 \times \frac{1}{2}} = 0.02\ \text{m} = 2\ \text{cm}$

S4. Let the shunt resistance be S.

Given: I = 750 A,

I_g = 100 A, R_G = 13 Ω



From the figure,

$I_g R_G = (I - I_g)S$

Or $100 \times 13 = [750 - 100]S$

Or $1300 - 650\ S$

$\therefore S = 1300/650 = 2\ \Omega$

S5. for L_1 $\oint \vec{B} \cdot \overrightarrow{d\ell} = m_0(I_1 - I_2)$

here I_1 is taken positive because magnetic lines of force produced by I_1 is anti-clockwise as seen from top. I_2 produces lines of $\vec{B}$ in clockwise sense as seen from top. The sense of $\overrightarrow{d\ell}$ is anticlockwise as seen from top.

for L_2 : $\oint \vec{B} \cdot \overrightarrow{d\ell} = \mu_0\ (I_1 - I_2 + I_4)$

for L_3 : $\oint \vec{B} \cdot \overrightarrow{d\ell} = 0$s

HOMEWORK EXERCISE SOLUTIONS

MCQ

S1. (b) 1 Gauss = 10^{-4} Tesla

S2. (b) Accelerate the positively charged particles
A cyclotron is a device used to accelerate positively charged particles.

S3. (d) $B = \mu_0 nI$
Magnetic field at any point inside the straight solenoid is given as $B = \mu_0 nI$.

S5. (c) Lorentz force
When the charged particles move in a combined magnetic and electric field, then the force acting is known as Lorentz force.

S6. (c) Tesla
The SI unit of the magnetic field is Tesla.

S7. (b) the magnetic moment of a current I carrying a circular coil of radius r and number of turns N varies as r^2.

S8. (d) F = -evB sin 180° = 0 (i.e 0= 0°or 180° in both cases F = 0). The electron will continue to move with uniform velocity or will go undeflected along the axis of the solenoid.

S9. (b) A current carrying closed loop of any shape when placed in a uniform magnetic field does not experience any force.

S10. (d) In non- uniform magnetic field current loop experiences an unequal opposite force which forms torque.

ASSERTION AND REASONING

S1. (c) Galvanometer to ammeter conversion takes place by connecting a low value resistance known as shunt in parallel with it. The assertion is true.
When two resistors are connected in parallel then the effective resistance becomes lowers. Hence the reason in false.

S2. (d) It is quite clear that magnetic poles always exist in pairs. Since, one can imagine magnetic field configuration with three poles. When north poles or south poles of two magnets are glued together. They provide a three-pole field configuration. It is also known that a bar magnet does not exert a torque on itself due to own its field.

S3. (c) In an atom, electrons revolve around the nucleus and as such the circular orbits of electrons may be considered as the small current loops. In addition to orbital motion, an electron has got spin motion also. So the total magnetic moment of electron is the vector sum of its magnetic moments due to orbital and spin motion. Charge particles at rest do not produce electric field.

S4. (d) The earth has only vertical component of its magnetic field at the magnetic poles. Since compass needle is only free to rotate in horizontal plane. At north pole the vertical component of earths field will exert torque on the magnetic needle so as to align it along its direction. As the compass needle cannot rotate in vertical plane, it will rest horizontally, when placed on the magnetic pole of the earth.

S5. (b) The susceptibility of ferromagnetic substance decreases with the rise of temperature in a complicated manner. After Curies point the susceptibility of ferromagnetic substance varies inversely with its absolute temperature. Ferromagnetic substance obeys Curies law only above its curie point.

VERY SHORT ANSWER

S1. Charge particle moves inclined to the magnetic field. When there is an angle between velocity of charged particle and magnetic field, then the vertical component of velocity (v sin θ) will rotate the charge particle on circular path, but horizontal component (v cos θ) will move the charged particle in straight line. Hence path of the charge particle becomes helical.

S2. Magnetic field lines can be entirely confined within the core of a toroid because toroid has no ends. A solenoid is open ended and the field lines inside it which are parallel to the length of the solenoid, cannot form closed curves inside the solenoid.

S3. Magnetic field is parallel or antiparallel to velocity of electron i.e., angle between v and B is 0° or 180°.

S4. By Fleming's left hand rule magnetic field must be along negative z-axis.

S5. An ammeter is connected in series with the circuit to read the current. If it had large resistance, it will change the current in circuit which it has to measure correctly; hence ammeter reading will have significant error; so for correct reading an ammeter should have a very low resistance.

NUMERICAL TYPE QUESTIONS

S1. $\vec{F} = q\vec{v} \times \vec{B} = 2 \times 10^{-6}\,(2\hat{\imath} - 3\hat{\jmath} \times 4\hat{k}) \times (3\hat{\jmath} - 2\hat{k}) = 2 \times 10^{-6}\,[-6\hat{\imath} + 4\hat{\jmath} + 6\hat{k}]$ N

By Newton's Law $\vec{a} = \frac{\vec{F}}{m} = \frac{2\times10^{-6}}{5\times10^{-6}}(-6\hat{\imath} + 4\hat{\jmath} + 6\hat{k})$
$= 0.8\,(-3\hat{\imath} + 2\hat{\jmath} + 3\hat{k})$ m/s^2

S2. Magnetic force $(\vec{F}_m)$ and magnetic field $(\vec{B})$ are ⊥ er to each other.

So, $\vec{a}_M$ and $\vec{B}$ will also be ⊥ er .

$$\therefore a_m \cdot \vec{B} = |\vec{a}_M||\vec{B}|\cos 90^\circ$$
$$(2\hat{\imath} + x\hat{\jmath}) \cdot (-3\hat{\mathbf{i}} + 2\hat{\jmath} - 4\hat{k}) = 0$$
$$2(-3) + (x \times 2) + [0 \times (-4)] = 0$$
$$2x = 6$$
$$x = 3$$

S3. (a) The field H is dependent of the material of the core, and is

H = nI = 1000 × 2.0 = 2 ×10^3 A/m.

(b) The magnetic field B is given by

B = μ_r μ_0 H

= 400 × 4p ×10^{-7} (N/A^2) × 2 × 10^3 (A/m) = 1.0 T

(c) Magnetization is given by

M = (B– μ_0 H)/ μ_0

= (μ_r μ_0 H–μ_0 H) / μ_0 = (μ_r – 1) H = 399 × H

» 8 × 10^5 A/m

(d) The magnetizing current I_M is the additional current that needs to be passed through the windings of the solenoid in the absence of the core which would give a B value as in the presence of the core. Thus B = μ_r n_0 (I + I_M). Using I = 2A, B = 1 T, we get I_M = 794 A.

S4. The magnetic field in the empty space enclosed by the windings of a toroid carrying a current i_0 is μ_0 n i_0 where n is the number of turns per unit length of the toroid and μ_0 is permeability of free space.

If the space is filled by a core of some material of permeability μ, then the field is given by

B = μ n i_0

But μ = $\mu_0 u_r$, where μ_r is the relative permeability of the core material. Thus,

B = μ_0 u_r ni_0 or $\mu_r = \frac{B}{\mu_0 n i_0}$

Here B = 2.5 T, i_0 = 0.7 A and n = $\frac{3000}{2\pi r}$ m^{-1}, where r is the mean radius of the toroid

(r = $\frac{11+12}{2}$ = 11.5 cm 11.5 × 10^{-2} m). Thus,

$$\mu = \frac{2.5}{(4\pi\times10^{-7})\times(3000/2\pi\times11.5\times10^{-2})\times0.7}$$
$$= \frac{2.5\times11.5\times10^{-2}}{2\times10^{-7}\times3000\times0.7}$$

μ_r = 684.5

S5. B = $\frac{\Phi}{A} = \frac{2.4\times10^{-5}Wb}{0.2\times10^{-4}m^2}$ =1.2 Wb/m^2 = 1.2 N A^{-1} m^{-1}.

The magnetizing field (or magnetic intensity) H is 1600 Am^{-1}. Therefore, the magnetic permeability is given by

μ = $\frac{B}{H} = \frac{1.2NA^{-1}m^{-1}}{1600Am^{-1}}$ = 7.5 × 10^{-4} N/A^2.

Now, from the relation μ = μ_0 (1 + χ_m), the susceptibility is given by $\chi_m = \frac{\mu}{\mu_0} - 1$.

We know that μ_0 = 4p × 10^{-7} N/A^2.

$$\chi_m = \frac{7.5\times10^{-4}}{4\times3.14\times10^{-7}} - 1 = 596.$$